Year 13 Maths - Further integration

Similar questions - Integration by parts, trapezium rule, areas

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Use the trapezium rule with 2 strips of equal width to show that $\int_0^4 \frac{1}{1+\sqrt{x}}\,\mathrm{d}x \approx 2\sqrt{2} - \frac{2}{3}$. **[4]**

(b) Use the substitution $u = \sqrt{x}$ to show that the exact value of the integral is $4 - 2\ln 3$. **[6]**

(c) Hence use part (a) to estimate $\ln 3$, giving your answer to 3 significant figures. **[2]**

2

(a) Use integration by parts to show that $\int x \ln x\,\mathrm{d}x = \frac{1}{2}x^2 \ln x - \frac{1}{4}x^2 + c$. **[4]**

(b) Hence find the exact area of the region bounded by $y = x\ln x$, the x-axis and the line $x = \mathrm{e}$. **[3]**

(c) Find the exact value of $\int_0^{\pi/2} x\cos 2x\,\mathrm{d}x$. **[4]**

Question 1

(a) $h = 2$; $y(0) = 1$, $y(2) = \frac{1}{1+\sqrt{2}} = \sqrt{2}-1$, $y(4) = \frac{1}{3}$. $T = \frac{2}{2}\left(1 + 2(\sqrt{2}-1) + \frac{1}{3}\right) = 2\sqrt{2} - \frac{2}{3}$

(b) $x = u^2$, $\mathrm{d}x = 2u\,\mathrm{d}u$, limits 0 to 2: $\displaystyle\int_0^2 \frac{2u}{1+u}\mathrm{d}u = \int_0^2 \left(2 - \frac{2}{1+u}\right)\mathrm{d}u = [2u - 2\ln(1+u)]_0^2 = 4 - 2\ln 3$

(c) $4 - 2\ln 3 \approx 2\sqrt{2} - \frac{2}{3}$, so $\ln 3 \approx 2 - \sqrt{2} + \frac{1}{3} = 0.919$

Question 2

(a) $u = \ln x$, $v' = x$: $\frac{1}{2}x^2 \ln x - \displaystyle\int \tfrac{1}{2}x^2 \cdot \tfrac{1}{x}\,\mathrm{d}x = \tfrac{1}{2}x^2 \ln x - \tfrac{1}{4}x^2 + c$

(b) The curve meets the axis at $x = 1$ and is positive for $1 < x \leqslant \mathrm{e}$. Area $= \left[\frac{1}{2}x^2 \ln x - \frac{1}{4}x^2\right]_1^{\mathrm{e}} = \left(\frac{\mathrm{e}^2}{2} - \frac{\mathrm{e}^2}{4}\right) + \frac{1}{4} = \frac{1}{4}(\mathrm{e}^2 + 1)$

(c) $\left[\frac{1}{2}x\sin 2x\right]_0^{\pi/2} - \displaystyle\int_0^{\pi/2} \tfrac{1}{2}\sin 2x\,\mathrm{d}x = 0 + \left[\tfrac{1}{4}\cos 2x\right]_0^{\pi/2} = -\tfrac{1}{4} - \tfrac{1}{4} = -\tfrac{1}{2}$